

Sriram Hariharan Neelakantan

Harrison, NJ | +1 (857) 437 9513 | sriramneel97@gmail.com | [LinkedIn](#)

PROFESSIONAL EXPERIENCE

Techmentee Inc. , Sr. Data Scientist | Harrison, NJ

Jul 2025 – Current

- Cut average response time from 3 min to 45 sec by developing an LLM-based multi-agent intent framework (LangGraph + Step Functions) automating tier-1 FinTech chats like balance and KYC queries
- Expanded automation to nine workflows by building API-driven Action Agents (Lambda + Step Functions) for refund, payment retry, and card block, removing human review for low-risk cases
- Boosted resolution accuracy by 12% via a human-in-the-loop feedback loop retraining classifier agents and refining Bedrock prompt logic with dynamic evaluation metrics

L.B. Foster Salient Systems, Data Scientist II | Dublin, OH

Dec 2024 – May 2025

- Built ETL pipelines to validate 3M+ vendor invoice and payment reconciliation records, improving financial data integrity and reducing downstream chargeback investigation delays by 30%
- Created standardized wheel health KPIs that enabled leadership to benchmark performance across fleets, leading to earlier issue detection and a 25% reduction in reactive maintenance cases
- Developed Python and SQL-based models and built dashboards in Power BI to visualize fraudulent vendor payments, contract compliance breaches, and predictive expense risks, improving case resolution speed by 20%
- Designed parameterized SQL queries and Python scripts to run A/B test scenarios on payment policies, including invoice thresholds and reimbursement rules, cutting false positives by 12% without raising fraud payment losses

Norfolk Southern Corp, Data Science Intern | Atlanta, GA

Sep 2022 – Dec 2022

- Consolidated and cleaned 90k+ operational and billing records in PostgreSQL and Python, removing 15% duplicate entries and providing a reliable foundation for risk policy evaluation and compliance testing
- Performed exploratory analysis on 30k+ case review and claims transactions, surfacing 50+ anomalies that strengthened manual review effectiveness and boosted risk signal reliability by 40%
- Built Tableau dashboards to track agent productivity KPIs, review turnaround times, and dispute outcomes, enabling leadership to measure policy efficiency and cut backlog by 18%
- Designed automated SQL and Python validation scripts to flag suspicious freight invoices, duplicate debits, and noncompliant payroll entries, improving fraud detection coverage and reducing investigation cycle times by 10 hours weekly

Accenture, Data Analyst II | India

Mar 2020 – Jul 2021

- Performed lapse and retention risk assessments using Python with gradient boosting and customer segmentation with SQL, uncovering drivers of early terminations that informed policy changes and cut early-lapse rates by 15%
- Partnered with compliance teams to evaluate chargeback case reviews, streamlining evidence submission and dispute handling processes, which cut manual review effort by 35% and recovered 18% more disputed funds
- Led ACH returns on 500k+ payment transactions, identifying high-risk merchant cohorts and implementing policy changes that cut chargeback ratios by 22% and improved overall transaction approval rates by 15%
- Created Python-based risk rules to cross-check application attributes (email domain, device ID, IP reputation), reducing fraudulent applications that slipped through manual review by 20%

- Implemented A/B experiments on payment fraud policy thresholds for ACH payroll deposits (transaction velocity, return windows), balancing false positives versus fraud exposure and reducing review queue backlog by 20% without raising fraud losses
- Engineered a payment fraud monitoring system integrating real-time anomaly signals (transaction velocity, duplicate debits, geo-location) that prevented \$400k in annual write-offs

Accenture, Data Analyst I | India

Jan 2018 – Mar 2020

- Designed and deployed fraud detection models using anomaly detection techniques, identifying 20% more suspicious claim patterns and mitigating financial exposure by an estimated ₹1.2M annually
- Conducted experiments and A/B tests to measure the impact of revised fraud and risk policies on conversion rates, win-loss ratios, and overall customer trust metrics, reducing manual review cycles by 15%
- Developed and automated rule-based validation scripts in Python to cross-verify insurance claims against underwriting guidelines and compliance thresholds, improving review accuracy and reducing manual processing time by 40%
- Developed KPIs and interactive dashboards to monitor fraud and compliance risk, enabling leadership to assess policy performance and make timely adjustments and resulting in a 20% faster response to emerging threats
- Implemented a rule-based fraud detection engine for application screening, curtailing fraudulent account openings by 20%

ACADEMIC PROJECTS

Strategic Analysis of Hospitality Data | EDA, Classification, Time Series Analysis

- Employed SVM, Logistic Regression, and XGBoost to achieve high accuracy and performance in analyzing Expedia hotel data, complemented by in-depth EDA focusing on Average Daily Rate and cancellation trends using ARIMA

Topic Modeling - News Data | USML, Clustering

- Achieved an impressive coherence value of 0.68 across 14 well-defined topics by utilizing advanced techniques, including LDA, LSA, K-means clustering, and NMF alongside TF-IDF vectorization

SKILLS

Programming Languages & Frameworks: Python (NumPy, Pandas, Scikit-Learn), R, PySpark, Java (REST APIs), Git, Bitbucket, JIRA, Linux / UNIX

Databases & Cloud Technologies: PostgreSQL, MySQL, MongoDB, Apache Spark, AWS EC2, AWS S3, Hadoop, Redshift

Risk & Fraud Analytics: Anomaly detection, A/B testing, Policy simulation, KPI design & monitoring, Predictive modeling (supervised/unsupervised ML), Hypothesis testing, Time-series forecasting (ARIMA)

Data Engineering & ETL: ETL pipeline design, Data processing & feature engineering, Data quality validation, Reconciliation & audit reporting, ACH payroll reporting, Alteryx

Data Visualization & Reporting: Tableau, Excel, Matplotlib, Seaborn, Power BI

EDUCATION

Master of Science, MS in Data Science

Sep 2021 – Dec 2023

Northeastern University, Boston, MA